

Experiment 'p_5_4_middle' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_4_middle

Natural language statement: null

Method signature: p_5_4_middle(str: string) returns (middle_str: string)

Ensures

- if |str| % 2 == 1 then |middle_str| == 1 else |middle_str| == 2
- if |str| % 2 == 1 then middle_str[0] == str[|str| / 2] else middle_str == str[|str| / 2 - 1 .. (|str| / 2) + 1]

Requires

- 0 < |str|

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_5_4_middle(str: string) returns (middle_str: string)`

The method should respect the following contract:

requires $0 < |str|$, ensures if $|str| \% 2 == 1$ then $|middle_str| == 1$ else $|middle_str| == 2$, ensures if $|str| \% 2 == 1$ then $middle_str[0] == str[|str| / 2]$ else $middle_str == str[|str| / 2 - 1 .. (|str| / 2) + 1]$

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnQuo7C7XhA8jeyp7UHL5wF5aRQAM

Model: gpt-5.5-2026-04-23

Created at: 1780673906

Finish reason: stop

Usage: 410 tokens in, and 674 tokens out

```
{
  var mid := |str| / 2;
  if |str| % 2 == 1 {
    middle_str := str[mid .. mid + 1];
    assert |middle_str| == 1;
    assert middle_str[0] == str[mid];
  } else {
    assert |str| % 2 == 0;
    assert 2 <= |str|;
    assert 1 <= mid;
    assert mid + 1 <= |str|;
    middle_str := str[mid - 1 .. mid + 1];
    assert |middle_str| == 2;
  }
}
```

Final Program

```

method p_5_4_middle(str: string) returns (middle_str: string)
  requires 0 < |str|
  ensures if |str| % 2 == 1 then |middle_str| == 1 else |middle_str| == 2
  ensures if |str| % 2 == 1 then middle_str[0] == str[|str| / 2] else middle_str ==
    ↪ str[|str| / 2 - 1 .. (|str| / 2) + 1]
{
  var mid := |str| / 2;
  if |str| % 2 == 1 {
    middle_str := str[mid .. mid + 1];
    assert |middle_str| == 1;
    assert middle_str[0] == str[mid];
  } else {
    assert |str| % 2 == 0;
    assert 2 <= |str|;
    assert 1 <= mid;
    assert mid + 1 <= |str|;
    middle_str := str[mid - 1 .. mid + 1];
    assert |middle_str| == 2;
  }
}

```

Total Token Usage

Input tokens: 410

Output tokens: 674

Reasoning tokens: 512

Sum of 'total tokens': 1084

Experiment Timings

Overall Experiment started at 1780673906254, ended at 1780673949102, lasting 42848ms (42.85 seconds)

Iteration #1 started at 1780673906261, ended at 1780673949102, lasting 42841ms (42.84 seconds)